

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
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Date:		Duration:	60 minutes

Code of Conduct

I S. Dhivya Sri / 192511246 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

1. Understand the Problem Statement

1.1 Problem Explanation

Agriculture involves continuous monitoring and timely decision-making related to crop growth, irrigation, weather conditions, fertilizer application, and day-to-day farm activities. In many farming environments, these activities are managed manually through memory, notebooks, phone calls, or separate information sources. This can lead to missed irrigation schedules, inappropriate fertilizer usage, delayed responses to weather changes, and poor activity tracking.

The proposed Smart Agriculture Management System is a centralized software application that brings these activities together in one platform. A farmer can maintain farm and crop information, receive weather updates, manage irrigation schedules, record agricultural activities, view fertilizer requirements, and receive reminders. The system supports better planning and informed agricultural decisions while keeping the farmer in control of final decisions.

1.2 Objectives

- Provide a centralized platform for managing farm and crop information.
- Help farmers plan and track irrigation schedules based on crop requirements.
- Provide accessible weather information to support agricultural planning.
- Maintain fertilizer requirements and application schedules.
- Record important agricultural activities and their completion status.
- Send timely notifications and reminders.
- Reduce dependence on scattered manual records.
- Support agricultural experts and administrators in managing relevant information.
- Improve farm activity monitoring, planning, and decision-making.

1.3 Scope

The scope covers a centralized web/mobile application for farmers, agricultural experts, administrators, and relevant information providers. It includes user management, farm and crop management, irrigation scheduling, weather information, fertilizer planning, activity tracking, notifications, and administration.

The initial system does not require fully automated farm machinery. IoT soil sensors, automatic irrigation, AI disease detection, drone monitoring, market-price integration, multilingual voice support, and government-service integration can be introduced as future enhancements.

2. Identify Key Stakeholders and Challenges

2.1 Stakeholders

Stakeholder	Role / Responsibility	Needs / Concerns
Farmers	Primary users; enter farm/crop details, manage schedules, record activities and view alerts.	Simple interface, reliable information, reminders and easy access.
Agricultural Experts	Provide crop, fertilizer, irrigation and farming guidance.	Accurate farm context and effective communication.
System Administrator	Manages users, data, access rights and configuration.	Security, monitoring, data quality and availability.
Weather Data Provider	Supplies current and forecast weather through an API/service.	Reliable data exchange and service availability.
Agricultural Input Providers	Provide fertilizer/product information where applicable.	Correct product information and integration.
Government / Agricultural Departments	May provide schemes, advisories and agricultural information.	Information dissemination and future integration.
Development & Maintenance Team	Designs, develops, tests, deploys and maintains software.	Clear requirements, maintainability and scalability.

2.2 Major Challenges

- Manual record keeping can result in missing, outdated or inconsistent information.
- Farmers may miss irrigation or fertilizer schedules because of workload or lack of reminders.
- Weather conditions can change rapidly and affect irrigation and farm activities.
- Incorrect fertilizer selection or timing can increase cost and negatively affect crops.
- Users may have different levels of digital literacy, requiring a simple interface.
- Internet connectivity may be unreliable in some agricultural locations.
- Sensitive user and farm information must be protected from unauthorized access.
- External weather services may experience downtime or provide incomplete data.
- Different crops and farms require different schedules and recommendations.

2.3 Stakeholder-Centred Analysis

The farmer is the primary stakeholder, so usability and practical value are critical. Agricultural experts contribute domain knowledge, while administrators ensure secure and reliable operation. External weather services provide important environmental information. The development team must convert these needs into testable requirements and maintain the application as requirements evolve.

3. Analyze the Current Approach

3.1 Existing System / Current Process

In a traditional approach, farmers may use notebooks, calendars, personal memory, phone calls, local agricultural experts, weather applications, and separate fertilizer information sources. Crop activities and irrigation are often tracked independently. Weather information may be checked manually from television, websites, mobile applications, or other sources.

This approach can work for small-scale operations because it is familiar and inexpensive, but it does not provide a unified view of farm activities. Information is distributed across multiple sources, making planning, reminders, and historical tracking difficult.

3.2 Strengths of Existing Approach

- Familiar to farmers and requires little technical training.
- Manual records can be maintained without specialized software.
- Local agricultural experts can provide personalized practical advice.
- Basic weather and agricultural information is widely available through existing channels.

3.3 Limitations and Gaps

Current Limitation	Impact / Gap
Scattered information	Farmers must switch between notebooks, apps, experts and other sources.
Manual scheduling	Important irrigation or fertilizer activities may be forgotten or delayed.
Limited centralized history	Past crop activities and applications are difficult to analyze.
Delayed weather awareness	Changes in weather may not be linked to planned farm activities.
Limited personalized guidance	Generic information may not match a specific crop or farm.
Connectivity dependence	Online features may become difficult to access during poor connectivity.
Security concerns	Manual or loosely managed records may lack proper access control.

3.4 Key Gap to Be Addressed

The major gap is the absence of a single, reliable platform that connects farm records, schedules, weather information, fertilizer requirements, agricultural activities and reminders. The proposed system addresses this gap through centralized data management and modular software services.

4. Propose a Suitable Solution Using Software Engineering Principles

4.1 Proposed Software Solution

The Smart Agriculture Management System is a centralized, modular application integrating farm information, crop management, irrigation scheduling, weather information, fertilizer planning, agricultural activity tracking, notifications and administration. Relevant information is stored in a structured database and presented through a user-friendly interface.

Role-based access ensures that farmers, experts and administrators receive appropriate functions. Weather information can be obtained through a secure API. The architecture should separate the presentation layer, application/business logic, data-access layer and external service integration so that individual components can be modified without affecting the entire system.

4.2 Major Functional Modules

Module	Main Functions
User Management	Registration/login, profile management, role-based access and authentication.
Farm & Crop Management	Add farms, fields, crop types, planting dates and crop status.
Irrigation Management	Create schedules, track completion and send reminders.
Weather Information	Display current/forecast weather and relevant alerts.
Fertilizer Management	Maintain fertilizer requirements, schedules and application records.
Agricultural Activity Tracking	Record sowing, irrigation, fertilization, pest-control and harvesting activities.
Notification & Reminder	Send timely alerts for scheduled and important activities.
Expert Guidance	Provide agricultural advice through authorized experts.
Admin Module	Manage users, system data, permissions and monitoring.
Reports & History	Provide activity history, crop records and farm summaries.

4.3 Proposed System Workflow

Farmer logs in → selects farm/field → enters crop details → checks weather → creates irrigation/fertilizer schedules → receives reminders → records completed activities → reviews history/reports. Experts can review relevant information and provide guidance, while administrators monitor users and system configuration.

4.4 Functional Requirements

- Secure registration and login.
- Role-based access for farmers, experts and administrators.
- Add, edit and view farm and crop details.
- Create, update and track irrigation schedules.
- Retrieve and display weather information.
- Maintain fertilizer requirements and application records.
- Record agricultural activities with date and status.
- Generate reminders and notifications.
- Maintain historical farm and activity records.
- Allow administrators to manage users and system information.
- Provide reports or summaries for monitoring activities.

4.5 Non-Functional Requirements

Requirement	Application in System
Usability	Simple navigation, clear labels, readable screens and farmer-friendly workflows.
Reliability	Validation, error handling, consistent storage and recovery mechanisms.
Security	Authentication, authorization and secure data handling.
Performance	Fast access to farm records and efficient schedule processing.
Scalability	Architecture and database can support increasing users and farms.
Maintainability	Modular code, documentation, testing and separation of concerns.
Availability	Suitable handling of external service failures and downtime.
Compatibility	Support for common browsers and supported mobile/web environments.
Data Integrity	Validation and controlled updates prevent inconsistent records.

4.6 Software Engineering Principles

Principle	Application
Modularity	Separate user, crop, irrigation, weather, fertilizer, activity, notification and admin modules.
Scalability	Layered architecture, efficient database design and API-based integration.
Maintainability	Clean code, documentation, reusable components, version control and testing.

Reliability	Input validation, exception handling, database consistency and backups.
Usability	Simple UI, consistent navigation and meaningful notifications.
Security	Secure authentication, role-based authorization and protected APIs.
Reusability	Common authentication, notification and validation services can be reused.
Testability	Independent modules and clear interfaces support unit and integration testing.

5. Justify the Chosen Methodology / Framework

5.1 Selected Methodology – Agile Scrum

Agile Scrum is selected because the Smart Agriculture Management System has multiple user groups and requirements that may evolve after feedback from farmers and agricultural experts. Weather integration, notification behaviour, crop-specific requirements and interface needs may require refinement during development.

Scrum divides development into short, time-boxed iterations called sprints. At the end of each sprint, a working increment is demonstrated, tested and reviewed. Feedback is incorporated into the product backlog and prioritized for later sprints.

5.2 Suggested Sprint Plan

Sprint	Main Deliverables
Sprint 1	Requirements, UI design, architecture, database design, registration and authentication.
Sprint 2	Farm and crop management, field details and crop records.
Sprint 3	Irrigation scheduling, activity tracking and reminders.
Sprint 4	Weather API integration, weather display and alerts.
Sprint 5	Fertilizer management, expert guidance and reporting.
Sprint 6	Security hardening, integration/usability testing, bug fixing and deployment preparation.

5.3 Why Agile Scrum Is Suitable

- Supports changing requirements through a prioritized product backlog.
- Provides frequent working software instead of waiting until project completion.
- Allows feedback from farmers and agricultural experts after each sprint.
- Encourages continuous testing, integration, review and improvement.
- Helps identify technical and usability problems early.
- Supports incremental delivery of high-priority functions such as irrigation and weather alerts.
- Improves communication through sprint planning, reviews and retrospectives.

5.4 Development and Maintenance

Agile supports maintenance because bugs, security updates, usability improvements and new features can remain in the product backlog after deployment. Changes can be prioritized and implemented in controlled sprints. Automated testing, version control, documentation, code reviews and continuous integration further improve software quality and maintainability.

6. Discuss Expected Outcomes and Limitations

6.1 Expected Benefits / Outcomes

- Centralized access to important farm and crop information.
- Better planning and tracking of irrigation and fertilizer activities.
- Timely reminders can reduce missed agricultural tasks.
- Weather information can support better farm scheduling and decision-making.
- Historical activity records improve monitoring and future planning.
- Experts can provide more context-aware guidance.
- Administrators can manage users and information systematically.
- Modular architecture provides a foundation for future IoT and AI integration.

6.2 Challenges Addressed

Challenge	Solution
Scattered information	Centralized farm, crop, weather, fertilizer and activity modules.
Missed schedules	Digital schedules, reminders and notifications.
Weather uncertainty	Integrated weather information and alerts.
Fertilizer planning	Crop-related fertilizer records and schedules.
Digital literacy	Simple, consistent and farmer-friendly interface.
Poor connectivity	Error handling and future offline/low-connectivity support.
Data security	Authentication, role-based access and secure data management.
Changing requirements	Agile Scrum, iterative feedback and backlog refinement.

6.3 Risks and Implementation Challenges

- Incorrect or incomplete weather data may affect decisions.
- External weather API failure may temporarily affect weather functions.
- Incorrect user-entered data may reduce recommendation quality.
- Unauthorized access may expose user or farm data if security is weak.
- Farmers may initially resist changing from familiar manual processes.
- Project scope may grow if advanced features are not prioritized.
- Future IoT integration can increase cost and technical complexity.

6.4 Limitations

- Depends on quality and availability of external weather information.
- Recommendations may not represent every local soil, climate or crop condition.
- Initial version cannot automatically measure soil moisture or control irrigation hardware without IoT.
- Some real-time features require internet connectivity.
- Accuracy of manually entered data depends on the user.

6.5 Future Enhancements

- IoT-based soil moisture, temperature and humidity monitoring.
- Automatic irrigation using sensors and smart controllers.
- AI-based crop disease and pest detection.
- AI-based crop, fertilizer and irrigation recommendations.
- Drone-based crop monitoring and image analysis.
- Market-price and agricultural-demand integration.
- Multilingual and voice-based interaction.
- Government agricultural scheme and advisory integration.
- Offline-first functionality for low-connectivity rural areas.
- Predictive analytics for crop yield and resource planning.

Conclusion

The Smart Agriculture Management System provides a practical software engineering solution for improving the planning and monitoring of agricultural activities. By centralizing crop, irrigation, weather, fertilizer and activity information, the system addresses major limitations of fragmented manual processes. Its modular architecture incorporates scalability, maintainability, reliability, usability and security, while Agile Scrum enables incremental development and continuous stakeholder feedback.

The system can begin as a manageable software application and progressively evolve through IoT, AI, automation, multilingual support and external agricultural-service integrations. Therefore, the proposed solution is feasible, user-oriented and suitable for systematic development using modern software engineering practices.

Criteria	Mark
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25